

**Course Title: Introduction to Programming**

**Course ID: CSE 06131209**

**Credit Hours: 3**

**Pre-requisite Courses: CSE 06131209**

**Continuous assessment: 60**

**Final exam: 40**

**Rationale:**

Introduces the fundamental concepts of structured programming, and provides a comprehensive introduction to programming for computer science and technology. Catalyzes the use of the C programming language as an aid to solving mathematical and scientific problems. Students design, write, and implement programs.

**Objectives:**

1. Ensure students grasp the syntax and semantics of the C programming language, including variables, data types, control structures, functions, and basic I/O operations.
2. Familiarize students with memory management concepts such as pointers, dynamic memory allocation, and memory deallocation to enable efficient utilization of memory resources.
3. Cultivate algorithmic thinking and problem-solving skills by presenting students with programming challenges and guiding them to develop efficient solutions using C programming constructs.
4. Teach students how to write modular and reusable code by organizing programs into functions, using header files, and adhering to best practices in code structure and organization.

**Semester and Year offered: Semester 2 Year 1**

**Student Learning Time(SLT)**

| Total Student Learning Time (SLT) | Face to Face<br>(hours)         | Total Guided and Independent Learning<br>(hours) |
|-----------------------------------|---------------------------------|--------------------------------------------------|
| Lecture (Offline)                 | 20<br>(1.40 hours x 12 weeks)   | 40<br>(3.20 hours x 12 weeks)                    |
| Lecture (Online)                  | 16.50<br>(1.30 x 11 weeks)      | 33<br>(3 hours x 11 weeks)                       |
| Tutorial (Review Class)           | 1.50<br>(1.30 x 1 week)         | 3<br>(3 hours x 1 week)                          |
| Forum Discussion                  | -                               | 6 hours<br>(12 Forum Discussion x 0.50 hours)    |
| Quiz                              | 0.50<br>(2 Quizzes × 0.25 hour) | 4<br>(2 Quizzes × 2 hours)                       |
| Assignment                        | -                               | 8<br>(4 Assignment x 2 hours)                    |
| Midterm                           | 1<br>(1 Exam × 1 hour)          | 2<br>(1 Exam × 2 hours)                          |
| Final Examination                 | 2<br>(1 Exam × 2 hours)         | 4<br>(1 Exam × 4 hours)                          |
| Subtotal                          | 41.50 hours                     | 100 hours                                        |
| Total (hours)                     | 141.50 hours                    |                                                  |

#### Course Outcomes (COs):

After completing this course students can:

1. Identify situations where computational methods and computers would be useful given a computational problem, identify and abstract the programming task involved.
2. Approach the programming tasks using techniques learned and write pseudo-code and choose the right data representation formats based on the requirements of the problem.
3. Use the comparisons and limitations of the various programming constructs and choose the right one for the task in hand.
4. Write the program on a computer, edit, compile, debug, correct, recompile and run it and identify tasks in which the numerical techniques learned are applicable and apply them to write programs, and hence use computers effectively to solve the task.

| Course Outcomes(COs)                                                                                                              | Course Content                                                                                                                                                                             | Teaching Learning Strategy                        | Assessment Strategy     |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|
| <b>CO1:</b> Identify situations where computational methods and computers are advantageous for solving problems, and abstract the | Introduction to the unit, fundamental concepts, first program<br>Numerical, Boolean and Character types; Variables and expressions<br>Variables: Assignment, Operators, Addressable memory | Lecture,<br>Discussion,<br>Self-Directed Learning | Assignment,<br>Mid Exam |

|                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                 |                                             |                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------|
| programming tasks involved.                                                                                                                                                             | Control Flow 1: Branching Control Flow 2: continue; basics of program design process, documentation and style<br>Addressable memory part 2. arrays, collections, Testing I Function<br>File input/output and Testing II<br>Programming idioms<br>Building programs from specification. Testing and Debugging<br>Recursion<br>Writing large programs. Structure and organisation |                                             |                        |
| <b>CO2:</b> Employ learned techniques to approach programming tasks, including writing pseudo-code and selecting appropriate data representation formats based on problem requirements. |                                                                                                                                                                                                                                                                                                                                                                                 | Lecture, Discussion, Self-Directed Learning | Assignment, Mid Exam   |
| <b>CO3:</b> Evaluate and contrast various programming constructs, considering their strengths and limitations, to select the most suitable for a given task.                            |                                                                                                                                                                                                                                                                                                                                                                                 | Lecture, Discussion, Self-Directed Learning | Assignment, Final Exam |
| <b>CO4:</b> Develop, debug, and refine programs on a computer, utilizing numerical techniques where applicable to effectively solve problems.                                           |                                                                                                                                                                                                                                                                                                                                                                                 | Lecture, Discussion, Self-Directed Learning | Assignment, Final Exam |

#### Text and Reference Books

1. Yashavant Kanetkar: Let us C-BPB Publications– 16th Edition (2017)
2. Herbert Schildt: Teach Yourself C – McGraw Hill – 3rd Edition (1997).
3. Balagurusamy E. programming in ANSI C. Tata McGraw-Hill Education; 2012.

#### COs-POs Mapping

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | Taxonomy       |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|----------------|
| CO1 | √   |     |     |     |     |     |     |     |     |      |      |      | C1, C2         |
| CO2 |     | √   |     |     |     |     |     |     |     |      |      |      | C2, C3, C4     |
| CO3 |     |     |     | √   |     |     |     |     |     |      |      |      | C3, C4, C5, A5 |
| CO4 |     |     | √   |     |     |     |     |     |     |      |      |      | C5, C6, P1, P3 |

#### COs -LODBNQF Mapping

|  |                    |               |                 |                 |
|--|--------------------|---------------|-----------------|-----------------|
|  | Fundamental Domain | Social Domain | Thinking Domain | Personal Domain |
|--|--------------------|---------------|-----------------|-----------------|

|            |   |  |  |  |
|------------|---|--|--|--|
| <b>CO1</b> | √ |  |  |  |
| <b>CO2</b> | √ |  |  |  |
| <b>CO3</b> | √ |  |  |  |
| <b>CO4</b> | √ |  |  |  |

\*LODBNQF=Learning Outcome Domain of Bangladesh National Qualification Framework

**Complex Engineering Problem(CP), Complex Activities(CA),Knowledge Profile(K) and Assessment Mapping with COs**

|            | Complex Engineering Problem(CP) | Complex Activities(CA) | Knowledge Profile(K) | Assessment             |
|------------|---------------------------------|------------------------|----------------------|------------------------|
| <b>CO1</b> | -                               | -                      | <b>K1</b>            | Assignment, Mid exam   |
| <b>CO2</b> | -                               | -                      | <b>K3</b>            | Assignment, Mid exam   |
| <b>CO3</b> | -                               | -                      | <b>K5</b>            | Assignment, Final Exam |
| <b>CO4</b> | -                               | -                      | <b>K8</b>            | Assignment, Final Exam |

**Bloom Taxonomy**

| Cognitive Domain (Knowledge-based) | Affective Domain (Emotive-based) | Psychomotor Domain (Action-based) |
|------------------------------------|----------------------------------|-----------------------------------|
| C1: Remembering                    | A1: Receiving                    | P1: Perception                    |
| C2: Understanding                  | A2: Responding                   | P2: Set                           |
| C3: Applying                       | A3: Valuing                      | P3: Guided Response               |
| C4: Analyzing                      | A4: Organizing                   | P4: Mechanism                     |
| C5: Evaluating                     | A5: Characterizing               | P5: Complex overt response        |
| C6: Synthesizing (Creating)        |                                  | P6: Adaptation                    |
|                                    |                                  | P7: Origination                   |

**Program Outcomes (POs)**

|                            |                           |                          |
|----------------------------|---------------------------|--------------------------|
| PO1: Engineering Knowledge | PO5: Modern Tools         | PO9: Communication       |
| PO2: Problem Analysis      | PO6: Engineer and Society | PO10: Team Work          |
| PO3: Design / Development  | PO7: Sustainability       | PO11: Lifelong Learning  |
| PO4: Investigation         | PO8: Ethics               | PO12: Project Management |

**Details Course Outlines**

| Week | Course Content                                                                                                                                                                                                                                         | Teaching Learning Strategy                           | Assessment Strategy                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------|
| 1    | Program Outcome (PO),<br>Vision, Mission, Objectives.<br>CO – PO relationship<br>Introduction to the unit, fundamental concepts, first program                                                                                                         | Lecture,<br>Discussion,<br>Self-Directed<br>Learning | Forum Discussion,<br>Class Participation,<br>Quiz, Mid exam,<br>Assignment   |
| 2    | Pseudocode writing                                                                                                                                                                                                                                     |                                                      |                                                                              |
| 3    | Numerical, Boolean, and Character types; Variables and expressions                                                                                                                                                                                     |                                                      |                                                                              |
| 4    | Variables: Assignment, Operators, Addressable memory                                                                                                                                                                                                   |                                                      |                                                                              |
| 5    | Control Flow 1: Branching Control                                                                                                                                                                                                                      |                                                      |                                                                              |
| 6    | Flow 2: continue; the basics of the program design process, documentation, and style                                                                                                                                                                   |                                                      |                                                                              |
| 7    | Develop code clichés for input and manipulating arrays, including maximum, minimum, search or traverse, with actions on each element for counting or summation                                                                                         |                                                      |                                                                              |
| 8    | Midterm Examination                                                                                                                                                                                                                                    |                                                      |                                                                              |
| 9    | Describe functionality to small procedural programs or functions.                                                                                                                                                                                      | Lecture,<br>Discussion,<br>Self-Directed<br>Learning | Forum Discussion,<br>Class Participation,<br>Quiz, Final Exam,<br>Assignment |
| 10   | File input/output and Testing II                                                                                                                                                                                                                       |                                                      |                                                                              |
| 11   | Justify read, understand, and trace procedural code.                                                                                                                                                                                                   |                                                      |                                                                              |
| 12   | Justify testing and debugging programs.                                                                                                                                                                                                                |                                                      |                                                                              |
| 13   | Perform to trace and write recursive code for evaluating recursively-defined numerical functions, and for recursively described array manipulations<br>Justify to write a structured algorithmic design to solve the descriptive problem specification |                                                      |                                                                              |
|      | Review Class                                                                                                                                                                                                                                           |                                                      |                                                                              |
| 14   | Final Examination                                                                                                                                                                                                                                      |                                                      |                                                                              |